

Note: This document only contains the graded problems viz Exercise 5. To view solution of entire Series 9 [click here](#).

Exercise 3

An instrument is used to measure the radii of circles. However, the instrument takes the measurements with an additional $\text{Normal}(0, \sigma^2)$ noise. Suppose n independent measurements $R_1, \dots, R_n$ of circles with the same radius r (unknown) are made. Our parameter of interest is $g(r) = \pi r^2$, the area of the circle.

- (a) Find the MLE of $g(r)$. Is it unbiased?
- (b) If the estimator in part (a) is biased, then find an unbiased estimator of $g(r)$. Is it the UMVUE?
- (c) If the estimator in part (b) is not the UMVUE, find the UMVUE.

Solution:

- (a) We are given that the instrument takes the measurement with an additional $\text{Normal}(0, \sigma^2)$ noise. Let R be the measured radius of such that $R = r + \varepsilon$ where $\varepsilon \sim N(0, \sigma^2)$ and r be the true radius. Thus, we have that

$$R_i \stackrel{iid}{\sim} N(r, \sigma^2) \forall 1 \leq i \leq n$$

Consider the following likelihood function

$$\begin{aligned} L(\sigma^2 | \underset{\sim}{R} = \underset{\sim}{r}) &= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma^2} (r_i - r)^2 \right\} \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (r_i - r)^2 \right\} \end{aligned}$$

Now, consider the log-likelihood function

$$l(r, \sigma^2) = L(\sigma^2 | \underset{\sim}{R} = \underset{\sim}{r}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (r_i - r)^2$$

Now, we'll compute the derivative for the log-likelihood w.r.t r and set it to 0 to obtain the MLE for r . Thus, we have the following

$$\frac{\partial l}{\partial r} = 0 \implies \frac{1}{\sigma^2} \sum_{i=1}^n (r_i - r) = 0 \implies r = \frac{1}{n} \sum_{i=1}^n r_i$$

Thus, we have that

$$\hat{r}_{ML} = \bar{R}_n$$

is the MLE for r . Since the maximum likelihood estimator is invariant, we know that if T is MLE of θ then $g(T)$ is MLE of $g(\theta)$. Thus, we have the following

$$A = g(r) = \pi r^2 \implies \hat{A}_{ML} = \pi \hat{r}^2$$

This implies that

$$\boxed{\hat{A}_{ML} = \pi (\bar{R}_n)^2}$$

Now, we'll check for unbiasedness. Since $R_i \stackrel{iid}{\sim} N(r, \sigma^2) \implies \bar{R}_n \sim N(r, \sigma^2/n)$. Thus, we know that $\mathbb{E}[\bar{R}_n] = r$ and $\mathbb{E}[(\bar{R}_n)^2] = \sigma^2/n + r^2$. This implies that

$$\begin{aligned} \mathbb{E}[\hat{A}_{ML}] &= \pi \mathbb{E}[(\bar{R}_n)^2] \\ &= \pi \left(\frac{\sigma^2}{n} + r^2 \right) = \pi r^2 + \frac{\pi \sigma^2}{n} \\ &= g(r) + \frac{\pi \sigma^2}{n} \end{aligned}$$

Thus, $\hat{A}_{ML}$ is biased with a positive bias of $\pi \sigma^2/n$

(b) Consider the following cases

- If σ^2 is known, consider the following estimator

$$T_1 = \hat{A}_{ML} - \frac{\pi \sigma^2}{n}$$

We can clearly see that $E(T_1) = g(r)$ thus making T_1 an unbiased estimator for $g(r)$. Also, if σ^2 is known then the given statistical model follows a single parameter normal distribution. In such case, we already know that $\bar{R}_n$ is complete and sufficient. Thus, following Lehmann-Scheffé theorem, we can establish that T_1 is UMVUE for $g(r)$

- If σ^2 is unknown, we'll simply replace σ^2 with its unbiased estimator S^2 . We know that

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (R_i - \bar{R}_n)^2 \sim \chi_{(n-1)}^2$$

We also know that $\mathbb{E}(S^2) = \sigma^2$ and S^2 is independent of $\bar{R}_n$ (by Basu's Theorem), thus, consider the following estimator

$$T_2 = \hat{A}_{ML} - \frac{\pi S^2}{n}$$

It is clearly evident that $\mathbb{E}(T_2) = g(r)$ thus making it truly unbiased for $g(r)$. We know that $(\bar{R}_n, S^2)$ is jointly complete and sufficient for (r, σ^2) . Thus following Lehmann-Scheffé theorem, we can establish that T_2 is UMVUE for $g(r)$

- (c) As stated in part (b), the derived unbiased estimators are UMVUE. Thus we have the following final result

$$T = \begin{cases} \pi (\bar{R}_n)^2 - \frac{\pi \sigma^2}{n} & \text{if } \sigma^2 \text{ is known} \\ \pi (\bar{R}_n)^2 - \frac{\pi S^2}{n} & \text{if } \sigma^2 \text{ is unknown} \end{cases}$$

is UMVUE for $g(r)$

Exercise 4

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta)$, $\theta \in [0, 1]$.

- For $n \geq 2$, show that $X_1 X_2$ is an unbiased estimator of θ^2 .
- Use part (a) to find the UMVUE of θ^2 .
- Recall that $\sum_{i=1}^n X_i \sim \text{Binomial}(n, \theta)$. Can you utilize this to derive the UMVUE?

Solution:

- (a) Given that $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(\theta)$, we know that $E[X_i] = \theta \forall 1 \leq i \leq n$ and also by virtue of independence that $E[X_1 X_2] = E[X_1] \cdot E[X_2]$. Thus, we have the following that

$$E[X_1 X_2] = E[X_1] \cdot E[X_2] = \theta^2$$

Thus, $T = X_1 X_2$ is unbiased for θ^2

- (b) To find a UMVUE for any given unbiased estimator of θ^2 , we'll do the *Rao-Blackwellisation* on the given unbiased estimator, T. For this, we need to find a complete and sufficient statistic, S of θ . Let $S = \sum X_i \sim \text{Bin}(n, \theta)$. We already know that S is sufficient, we'll show that S is complete.

Showing that S is complete

Consider $E[g(S)] = 0$ for some function g . We need to show that $P(g(s) = 0) = 1$. Consider the following expression

$$\begin{aligned} E[g(S)] &= \sum_{s=0}^n g(s) \binom{n}{s} \theta^s (1-\theta)^{n-s} = 0 \\ &= (1-\theta)^n \sum_{s=0}^n g(s) \underbrace{\binom{n}{s}}_{h(s)} \underbrace{\left(\frac{\theta}{1-\theta}\right)^s}_{\eta^s} = 0 \\ &\implies \sum_{s=0}^n h(s) \eta^s = 0 \end{aligned}$$

Since above equation forms a polynomial in η of degree n, thus by Fundamental Theorem of Algebra, it must have exactly n complex roots counting multiplicity

but since this is satisfied for all $\eta \in (0, \infty)$ it means that the above polynomial equation is an identity. Thus, all its coefficients are zero. This implies that

$$h(s) = 0 \implies g(s) = 0 \forall 0 \leq s \leq n$$

This proves that S is complete as well as sufficient. ■

Now, consider $T^* = \mathbb{E}[T|S = s]$. The Rao-Blackwell Theorem and Lehmann-Scheffé Theorem tell us that the obtained estimator, T^* will be the UMVUE for $\tau(\theta) = \theta^2$. Consider the following

$$X_1 X_2 = \begin{cases} 1 & \text{if } X_1 = X_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

This implies that,

$$\begin{aligned} \mathbb{E}[X_1 X_2 | S = s] &= P(X_1 = 1, X_2 = 1 | S = s) \\ &= \frac{P(X_1 = 1, X_2 = 1, \sum_{i=3}^n X_i = s - 2 | \sum_{i=3}^n X_i \sim \text{Bin}(n-2, \theta))}{P(\sum_{i=1}^n X_i = s | \sum_{i=1}^n X_i \sim \text{Bin}(n, \theta))} \\ &= \frac{P(X_1 = 1) \cdot P(X_2 = 1) \cdot P(\sum_{i=3}^n X_i = s - 2)}{P(\sum_{i=1}^n X_i = s)} \\ &= \frac{\theta^2 \binom{n-2}{s-2} (\theta)^{s-2} (1-\theta)^{n-s}}{\binom{n}{s} \theta^s (1-\theta)^{n-s}} \\ &= \frac{\binom{n-2}{s-2}}{\binom{n}{s}} = \frac{s(s-1)}{n(n-1)} \end{aligned}$$

This implies that

$$T^* = \frac{S(S-1)}{n(n-1)} = \frac{\sum X_i (\sum X_i - 1)}{n(n-1)}$$

is UMVUE for $\tau(\theta) = \theta^2$

- (c) As already derived in part (b), $S = \sum_{i=1}^n X_i \sim \text{Bin}(n, \theta)$ is complete and sufficient statistic for θ . Consider the following

$$\begin{aligned} \mathbb{E}[S(S-1)] &= \mathbb{E}[S^2] - \mathbb{E}[S] = \text{Var}(S) + (\mathbb{E}[S])^2 - \mathbb{E}[S] \\ &= n\theta(1-\theta) + n^2\theta^2 - n\theta = \theta^2(n^2 - n) \\ &= \theta^2 n(n-1) \end{aligned}$$

This implies that

$$\mathbb{E} \left[\underbrace{\frac{S(S-1)}{n(n-1)}}_{T^*} \right] = \theta^2$$

Since T^* is a function of the complete and sufficient statistic and is also unbiased, thus according to Lehmann-Scheffé Theorem, it is UMVUE